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By Amit Katiyar (MCA-JNU)



- The tangent to a curve $f(x)$ makes an angle $\pi/6$ at $x = 1$ and $\pi/4$ at $x = 4$ with the x-axis. The value of $\int_1^4 f'(x)f''(x) dx$ is:
NIMCET 2007
(a) $\frac{\sqrt{3}+1}{3}$ (b) $\frac{\sqrt{3}-1}{3}$ (c) 3 (d) $1/3$
- The value of $\int_{-2}^3 |1-x^2| dx$ is
NIMCET 2007
(a) $7/3$ (b) $14/3$ (c) $28/3$ (d) $1/3$
- $\int \frac{dx}{\cos x - \sin x}$ is equal to
NIMCET 2007
(a) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{3\pi}{8} \right) \right| + c$ (b) $\frac{1}{\sqrt{2}} \log \left| \cot \left(\frac{x}{2} \right) \right| + c$
(c) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{\pi}{8} \right) \right| + c$ (d) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right| + c$
- The value of $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x}$ is
NIMCET 2008
(a) 0 (b) 1 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$
- The value of $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$ is
NIMCET 2009
(a) $\frac{\pi^2}{3}$ (b) $\frac{\pi^2}{4}$ (c) $\frac{\pi^2}{6}$ (d) $\frac{\pi^2}{2}$
- The value of integral $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}} dx$ is
NIMCET 2010
(a) 1 (b) $1/2$ (c) $3/2$ (d) 2
- The value of integral $\int_0^{\pi/4} \frac{\sin x + \cos x}{3+\sin 2x} dx$ is
NIMCET 2010
(a) $\log 2$ (b) $\log 3$ (c) $1/4 \log 3$ (d) $1/8 \log 3$
- If $I_1 = \int_0^1 2x^2 dx$, $I_2 = \int_0^1 2x^3 dx$, $I_3 = \int_1^2 2x^2 dx$ and $I_4 = \int_1^2 2x^3 dx$, then
NIMCET 2010
(a) $I_3 = I_4$ (b) $I_3 > I_4$ (c) $I_2 > I_1$ (d) $I_1 > I_2$
- $\int_0^{1/2} \frac{dx}{\sqrt{x-x^2}}$ will be equal to
NIMCET 2011
(a) $\frac{\pi}{2}$ (b) π (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{4}$
- If 'a' is a positive integer, then the number of values satisfying $\int_0^{\pi/2} \left\{ a^2 \left(\frac{\cos 3x}{4} + \frac{3}{4} \cos x \right) + a \sin x - 20 \cos x \right\} dx \leq \frac{-a^2}{3}$
NIMCET 2011
(a) only one (b) two (c) three (d) four
- If $I_1 = \int_0^1 2x^2 dx$, $I_2 = \int_0^1 2x^3 dx$, $I_3 = \int_1^2 2x^2 dx$ and $I_4 = \int_1^2 2x^3 dx$, then
NIMCET 2012
(a) $I_1 = I_2$ (b) $I_2 > I_1$ (c) $I_3 > I_4$ (d) $I_4 > I_3$
- The value of integral $\int_0^{\pi/2} \log \tan x dx$ is
NIMCET 2012
(a) π (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) 0
- The value of $\int_0^{\sin^{-1} x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^{-1} x} \cos^{-1} \sqrt{t} dt$ is
NIMCET 2012
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) 1 (d) None
- If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$, then $I_8 + I_6$ is equal to
NIMCET 2013
(a) $1/4$ (b) $1/5$ (c) $1/6$ (d) $1/7$
- The value of the integral $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ is
NIMCET 2013
(a) 0 (b) $-\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$
- The value of $\int_0^{\pi/4} \log(1 + \tan x) dx$ is equal to
NIMCET 2014
(a) $\frac{\pi}{4} \log 2$ (b) $\frac{\pi}{6} \log 2$ (c) $\frac{\pi}{8} \log 2$ (d) $\frac{\pi}{2} \log 2$
- If $[x]$ represent the greatest integer not exceeding x, then $\int_0^9 [x] dx$
NIMCET 2014
(a) 32 (b) 36 (c) 40 (d) 28

- $\int_0^{\pi} [\cot x] dx$ where $[.]$ denotes the greatest integer function, is equal to
NIMCET 2015
(a) $\frac{\pi}{2}$ (b) 1 (c) -1 (d) $-\frac{\pi}{2}$
- The value of $\int_{-\pi/3}^{\pi/3} \frac{x \sin x}{\cos^2 x} dx$ is
NIMCET 2015
(a) $\frac{1}{3} (4\pi + 1)$ (b) $\frac{4\pi}{3} - 2 \log \tan \frac{5\pi}{12}$
(c) $\frac{4\pi}{3} + \log \tan \frac{5\pi}{12}$ (d) $\frac{4\pi}{3} - \log \tan \frac{5\pi}{12}$
- The value of $\int_0^{\pi} x^3 \sin x dx$ is
NIMCET 2017
(a) $\pi^3 - 6\pi$ (b) $-\pi^3 - 6\pi$
(c) $-\pi^3 + 6\pi$ (d) $\pi^3 + 6\pi$
- Evaluate $\int_0^1 x(1-x)^n dx$
NIMCET 2017
(a) $\frac{-1}{(n+1)(n+2)}$ (b) $\frac{1}{(n+1)(n+2)}$
(c) $(n+1)(n+2)$ (d) $(n-1)(n-2)$
- If $f(x) = \sin^5 x + \sin^3 x$ and $g(x) = \cos^5 x + \sin^3 x$, then the value of $\int_0^{\pi} [f(x) + f(-x)] [g(x) + g(-x)] dx$ is
NIMCET 2018
(a) 0 (b) > 1 (c) 0 and 1 (d) less than 0
- $\int_0^{\pi} x f(\sin x) dx$ is equal to
NIMCET 2018
(a) $\pi \int_0^{\pi} f(\sin x) dx$ (b) $\frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$
(c) $\pi \int_0^{\pi/2} f(\cos x) dx$ (d) $\pi \int_0^{\pi} f(\cos x) dx$
- Let $f: R \rightarrow R$ be defined by
 $f(x) = \begin{cases} x+2, & \text{if } x < 0 \\ |x-2|, & \text{if } x \geq 0 \end{cases}$ Find $\int_{-2}^3 f(x) dx$.
NIMCET 2018
(a) 0.5 (b) 2.5 (c) 4.5 (d) 6.5
- If $\int_{\log 2}^x \frac{1}{\sqrt{x-1}} dx = \frac{\pi}{6}$, then $x =$
NIMCET 2019
(a) $\log 2$ (b) $2 \log 2$ (c) $3 \log 2$ (d) $4 \log 2$
- If $I_n = \int_0^a (a^2 - x^2)^n dx$ where n is a positive integer, then the relation between I_n and I_{n-1} is
NIMCET 2020
(a) $I_n = \frac{2na^2}{2n+1} I_{n-1}$ (b) $I_n = \frac{2n^2 a^2}{2n+1} I_{n-1}$
(c) $I_n = \frac{2na^2}{2n-1} I_{n-1}$ (d) $I_n = \frac{2n^2 a^2}{2n-1} I_{n-1}$
- The value of $\int_{-1}^2 (ax^5 + bx^3 + c) dx$ depends on the
NIMCET 2020
(a) value of b (b) value of c
(c) value of a (d) value of a and b
- If $a < b$, then $\int_a^b (|x-a| + |x-b|) dx$, is equal to
NIMCET 2022
(a) $\frac{(b-a)^2}{2}$ (b) $\frac{(b^2-a^2)}{2}$ (c) $\frac{(b^3-a^3)}{2}$ (d) $(b-a)^2$
- For $a \in R$ (the set of all real numbers), $a \neq 1$,
 $\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$. Then one of the values of a is
NIMCET 2022
(a) 5 (b) 8 (c) $-15/2$ (d) $-17/2$
- The value of $\int_{-\pi/3}^{\pi/3} \frac{x \sin x}{\cos^2 x} dx$ is
NIMCET 2023
(a) $\frac{1}{3} (4\pi + 1)$ (b) $\frac{4\pi}{3} - 2 \log \tan \frac{5\pi}{12}$
(c) $\frac{4\pi}{3} + \log \tan \frac{5\pi}{12}$ (d) $\frac{4\pi}{3} - \log \tan \frac{5\pi}{3}$



31. The value of the integral $\int_0^{\frac{\pi}{2}} \frac{(1+2\cos x)}{(2+\cos x)^2} dx$ lies in the interval **NIMCET 2025**

- (a) (-2, -1) (b) (-1, 0)
(c) (0, 1) (d) (1, 2)

32. The value of $\frac{d}{dx} \int_{\sin x}^{2\sin x} e^{t^2} dt$ at $x = \pi$ is **NIMCET 2025**

- (a) -1 (b) 1
(c) -2 (d) 2

33. Which of the following is TRUE? (Integration)

NIMCET 2024

- (a) If f is continuous on $[a, b]$, then $\int_a^b xf(x) = x \int_a^b f(x) dx$
(b) $\int_0^3 e^{x^2} dx = \int_0^5 e^{x^2} dx + \int_5^3 e^{x^2} dx$
(c) If f is continuous on $[a, b]$, then $\frac{d}{dx} \left(\int_a^b f(x) dx \right) = f(x)$
(d) Both (a) and (b)

34. If for non-zero x , $cf(x) + df(1/x) = |\log|x|| + 3$, where $c \neq d$, then $\int_1^e f(x) dx =$ (Integration) **NIMCET 2024**

- (a) $\frac{(c-d)(2e-1)}{c^2-d^2}$ (b) $\frac{(c-d)(3e-2)}{c^2-d^2}$
(c) $\frac{(c-d)(3e+2)}{c^2-d^2}$ (d) $\frac{(c-d)(2e+1)}{c^2-d^2}$

ANSWER KEYS

1.	d	2.	c	3.	d	4.	c	5.	b
6.	c	7.	c	8.	d	9.	a	10.	d
11.	d	12.	d	13.	a	14.	d	15.	d
16.	c	17.	b	18.	d	19.	b	20.	a
21.	b	22.	a	23.	b	24.	c	25.	b
26.	a	27.	b	28.	d	29.	d	30.	b
31.	d	32.	c	33.	b	34.	b		

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	03	NIMCET 2016	00
NIMCET 2008	01	NIMCET 2017	02
NIMCET 2009	01	NIMCET 2018	03
NIMCET 2010	03	NIMCET 2019	01
NIMCET 2011	02	NIMCET 2020	02
NIMCET 2012	03	NIMCET 2021	00
NIMCET 2013	02	NIMCET 2022	02
NIMCET 2014	02	NIMCET 2023	01
NIMCET 2015	02	NIMCET 2024	00
		NIMCET 2025	01

SOLUTION

1. (d)

$$f'(1) = \tan \frac{\pi}{6}$$

$$f'(4) = \tan \frac{\pi}{4}$$

$$\int_1^4 f'(x) \cdot f''(x) dx$$

$$f'(x) = t$$

$$f''(x) dx = dt$$

$$\int_1^4 t dt \Rightarrow \left[\frac{t^2}{2} \right]_1^4$$

$$= \frac{1}{2} [(f'(x))^2]_1^4$$

$$= \frac{1}{2} [(f'(4))^2 - (f'(1))^2]$$

$$\frac{1}{2} \left[\left(\tan \frac{\pi}{4} \right)^2 - \left(\tan \frac{\pi}{6} \right)^2 \right]$$

$$\frac{1}{2} \left[1 - \frac{1}{3} \right]$$

$$\frac{1}{2} \left[\frac{2}{3} \right] = \frac{1}{3}$$

2. (c) $\int_{-2}^3 |1-x^2| dx$

$$\int_{-2}^{-1} -(1-x^2) dx + \int_{-1}^1 (1-x^2) dx + \int_1^3 -(1-x^2) dx$$

$$\left[\frac{x^3}{3} - x \right]_{-2}^{-1} + \left[x - \frac{x^3}{3} \right]_{-1}^1 + \left[\frac{x^3}{3} - x \right]_1^3$$

$$-\frac{1}{3} + 1 + \frac{2}{3} - 2 + 1 - \frac{1}{3} + 1 - \frac{1}{3} + 9 - 3 - \frac{1}{3} + 1$$

$$7 + \frac{7}{3} = \frac{28}{3}$$

3. (d) $\int \frac{dx}{\cos x - \sin x}$ is equal to

$$\frac{1}{\sqrt{2}} \int \frac{\frac{dx}{\sqrt{2}}}{\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x}$$

$$\frac{1}{\sqrt{2}} \int \frac{\frac{dx}{\sqrt{2}}}{\cos \frac{\pi}{4} \cos x - \sin \frac{\pi}{4} \sin x}$$

$$\frac{1}{\sqrt{2}} \int \frac{\frac{dx}{\sqrt{2}}}{\cos \left(\frac{\pi}{4} + x \right)}$$

$$\frac{1}{\sqrt{2}} \int \sec \left(\frac{\pi}{4} + x \right) dx$$

$$\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{\pi}{8} + \frac{\pi}{4} \right) \right| \Rightarrow \frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right|$$

$$= \frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} - \frac{\pi}{2} + \frac{\pi}{2} \right) \right| \Rightarrow \frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right| + C$$

4. (c) $I = \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan^2 x} = \int_0^{\frac{\pi}{2}} \frac{dx}{1+\frac{\sin^2 x}{\cos^2 x}}$

$$I = \int_0^{\pi/2} \frac{\cos^2 x}{\cos^2 x + \sin^2 x} dx$$

$$I = \int_0^{\pi/2} \frac{\cos^2 \left(\frac{\pi}{2} - x \right)}{\cos^2 \left(\frac{\pi}{2} - x \right) + \sin^2 \left(\frac{\pi}{2} - x \right)} dx$$

$$I = \int_0^{\pi/2} \frac{\sin^2 x}{\sin^2 x + \cos^2 x} dx$$

Add Eqs. (i) and (ii),

$$2I = \int_0^{\pi/2} \frac{\cos^2 x + \sin^2 x}{\sin^2 x + \cos^2 x} dx$$

$$2I = [x]_0^{\pi/2} \Rightarrow 2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

Or using shortcut formula

$$\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx = \frac{b-a}{2}$$

5. (b) $I = \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$

$$I = \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x)}{1+\cos^2(\pi-x)} dx = \int_0^{\pi} \frac{(\pi-x) \sin x}{1+\cos^2 x} dx$$

Add Eqs. (i) and (ii),

$$2I = \int_0^{\pi} \frac{\pi \sin x - x \sin x + x \sin x}{1+\cos^2 x} dx \Rightarrow 2I = \int_0^{\pi} \frac{\pi \sin x}{1+\cos^2 x} dx$$

Let $\cos x = t$

$$\Rightarrow -\sin x dx = dt \Rightarrow \sin x dx = -dt$$

$$2I = \pi \int_1^{-1} \frac{-dt}{1+t^2} \Rightarrow 2I = \pi \int_{-1}^1 \frac{dt}{1+t^2}$$

$$2I = \pi [\tan^{-1} t]_{-1}^1$$

$$2I = \pi [\tan^{-1} 1 - \tan^{-1}(-1)] = \pi \left[\frac{\pi}{4} + \frac{\pi}{4} \right]$$

$$2I = \pi \left[\frac{\pi}{2} \right] \Rightarrow I = \frac{\pi^2}{4}$$



6. (c) $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}} dx$

Using shortcut formula $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx$

$$= \frac{b-a}{2} = \frac{6-3}{2} = \frac{3}{2}$$

7. (c) $\int_0^{\pi/4} \frac{\sin x + \cos x}{3 + \sin 2x} dx$

$$\int_0^{\pi/4} \frac{\sin x + \cos x}{4 - 1 + \sin 2x} dx = \int_0^{\pi/4} \frac{\sin x + \cos x}{4 - (1 - \sin 2x)} dx$$

$$= \int_0^{\pi/4} \frac{(\sin x + \cos x) dx}{4 - (\cos^2 x + \sin^2 x - 2 \sin x \cos x)}$$

$$= \int_0^{\pi/4} \frac{(\sin x + \cos x) dx}{4 - (\sin x - \cos x)^2}$$

Let $\sin x - \cos x = t \Rightarrow (\cos x + \sin x) dx = dt$

$$= \int_{-1}^0 \frac{dt}{4-t^2} = \int_{-1}^0 \frac{dt}{2^2-t^2} = \frac{1}{2 \times 2} \left[\log \left| \frac{t+2}{t-2} \right| \right]_{-1}^0$$

$$= \frac{1}{4} \left[\log \left| \frac{0+2}{0-2} \right| - \log \left| \frac{-1+2}{-1-2} \right| \right]$$

$$= \frac{1}{4} \log 3.$$

8. (d) $I_1 = \int_0^1 2x^2 dx, I_2 = \int_0^1 2x^3 dx, I_3 = \int_1^2 2x^2 dx$

$$I_4 = \int_1^2 2x^3 dx$$

$$I_1 = \int_0^1 2x^2 dx, I_2 = \int_{10}^1 2x^3 dx$$

$$0 < x < 1$$

$$x^2 > x^3$$

$$2x^2 > 2x^3$$

$$\int_0^1 2x^2 dx > \int_0^1 2x^3 dx$$

$$I_1 > I_2$$

9. (a) $\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{x-x^2}}$

$$\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{-(x^2-x)}} = \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{-\left(x-\frac{1}{2}\right)^2 + \frac{1}{4}}}$$

$$= \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{\frac{1}{4} - \left(x-\frac{1}{2}\right)^2}} = \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{\left(\frac{1}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2}}$$

$$= \left[\sin^{-1} \left(\frac{x-\frac{1}{2}}{\frac{1}{2}} \right) \right]_0^{\frac{1}{2}} = \left[\sin^{-1} \left(\frac{\frac{1}{2}-\frac{1}{2}}{\frac{1}{2}} \right) \right]$$

$$= [\sin^{-1}(0) - \sin^{-1}(-1)] = \frac{\pi}{2}$$

10. (d) $\int_0^{\pi/2} \left\{ a^2 \left(\frac{\cos 3x}{4} + \frac{3}{4} \cos x \right) + a \sin x - 20 \cos x \right\} dx$

$$dx \leq -\frac{a^2}{3}$$

$$= \frac{a^2}{4} \int_0^{\pi/2} \cos 3x dx + \frac{3a^2}{4} \int_0^{\pi/2} \cos x dx$$

$$+ a \int_0^{\pi/2} \sin x dx - 20 \int_0^{\pi/2} \cos x dx \leq \frac{-a^2}{3}$$

$$= \frac{a^2}{4} \left[\frac{\sin 3x}{3} \right]_0^{\pi/2} + \frac{3a^2}{4} [\sin x]_0^{\pi/2}$$

$$+ a[-\cos x]_0^{\pi/2} - 20[\sin x]_0^{\pi/2} \leq \frac{-a^2}{3}$$

$$= \frac{a^2}{12} \left[\sin \frac{3\pi}{2} - \sin 0 \right]$$

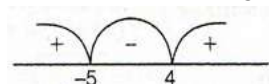
$$+ \frac{3a^2}{4} \left[\sin \frac{\pi}{2} - \sin 0 \right] + a \left[-(\cos \frac{\pi}{2} - \cos 0) \right]$$

$$- 20 \left[\sin \frac{\pi}{2} - \sin 0 \right] \leq \frac{-a^2}{3}$$

$$= \frac{a^2}{12} [-1] + \frac{3a^2}{4} [1] + a[1] - 20 \leq \frac{-a^2}{3}$$

Multiply by 12 on both sides.

$$8a^2 + 12a - 240 \leq \frac{-12a^2}{3}$$



$$\Rightarrow a^2 + a - 20 \leq 0 \Rightarrow (a+5)(a-4) \leq 0$$

Hence

$$-5 \leq a \leq 4$$

As a is a natural number, so $a \in \{1, 2, 3, 4\}$.

11. (d) $I_1 = \int_0^1 2x^2 dx, I_2 = \int_0^1 2x^3 dx, I_3 = \int_1^2 2x^2 dx$

$$\text{and } I_4 = \int_1^2 2x^3 dx$$

$$0 < x < 1 \Rightarrow x^2 > x^3$$

$$\int_0^1 2x^2 > \int_0^1 2x^3 \Rightarrow I_1 > I_2$$

$$1 < x < 2 \Rightarrow x^3 > x^2$$

$$\int_1^2 2x^3 > \int_1^2 2x^2 \Rightarrow I_4 > I_3$$

12. (d) $I = \int_0^{\pi/2} \log \tan x dx$

$$I = \int_0^{\pi/2} \log \tan \left(\frac{\pi}{2} - x \right) dx$$

$$I = \int_0^{\pi/2} \log \cot x dx$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi/2} (\log \tan x + \log \cot x) dx$$

$$= \int_0^{\pi/2} \log \tan x \times \cot x dx = \int_0^{\pi/2} (\log 1) dx$$

13. (a) $I = \int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt$

Check it for $x = \frac{\pi}{4}$

$$\int_0^{\frac{1}{2}} \sin^{-1} \sqrt{t} dt + \int_0^{\frac{1}{2}} \cos^{-1} \sqrt{t} dt$$

$$= \int_0^{\frac{1}{2}} [(\sin^{-1} \sqrt{t}) + (\cos^{-1} \sqrt{t})] dt = \int_0^{\frac{1}{2}} \frac{\pi}{2} dt = \frac{\pi}{4}$$

14. (d) $I_8 = \int_0^{\pi/4} \tan^8 \theta d\theta, I_6 = \int_0^{\pi/4} \tan^6 \theta d\theta$

$$I_8 + I_6 = \int_0^{\pi/4} \tan^8 \theta d\theta + \int_0^{\pi/4} \tan^6 \theta d\theta$$

$$= \int_0^{\pi/4} \tan^6 \theta (1 + \tan^2 \theta) d\theta = \int_0^{\pi/4} \tan^6 \theta \sec^2 \theta d\theta$$

$$\text{Let } \tan \theta = t \Rightarrow \sec^2 \theta d\theta = dt = \int_0^1 t^6 dt$$

$$I_8 + I_6 = \left[\frac{t^7}{7} \right]_0^1 = \frac{1}{7} - 0 = \frac{1}{7}$$

15. (d) $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

Using shortcut formula $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx = \frac{b-a}{2}$

$$\Rightarrow I = \frac{\frac{\pi}{2}-0}{2} \Rightarrow I = \frac{\pi}{4}$$

16. (c) $I = \int_0^{\pi/4} \log(1 + \tan x) dx$

$$I = \int_0^{\pi/4} \log \left(1 + \tan \left(\frac{\pi}{4} - x \right) \right) dx$$

$$I = \int_0^{\pi/4} \log \left(1 + \frac{1 - \tan x}{1 + \tan x} \right) dx$$

$$I = \int_0^{\pi/4} \log \left(\frac{2}{1 + \tan x} \right) dx$$

Add Eqs. (i) and (ii),



$$2I = \int_0^{\pi/4} \left[\log(1 + \tan x) + \log\left(\frac{2}{1+\tan x}\right) \right] dx$$

$$= \int_0^{\pi/4} \log \left[(1 + \tan x) \times \left(\frac{2}{1+\tan x}\right) \right] dx$$

$$2I = \int_0^{\pi/4} \log 2 dx \Rightarrow 2I = \left(\frac{\pi}{4}\right) \log 2 \Rightarrow I = \frac{\pi}{8} \log 2$$

17. (b) $\int_0^9 [x] dx$

$$\text{GIF} = \int_0^1 [x] dx + \int_1^2 [x] dx + \int_2^3 [x] dx$$

$$+ \int_3^4 [x] dx + \int_4^5 [x] dx + \int_5^6 [x] dx$$

$$+ \int_6^7 [x] dx + \int_7^8 [x] dx + \int_8^9 [x] dx$$

$$= \int_0^1 0 dx + \int_1^2 1 dx + \int_2^3 2 dx + \dots + \int_8^9 8 dx$$

$$= 0 + [x]_1^2 + 2[x]_2^3 + 3[x]_3^4 \dots + 8[x]_8^9$$

$$= 2 - 1 + 2(3 - 2) + 3(4 - 3) \dots 8(9 - 8)$$

$$= 1 + 2 + 3 \dots + 8$$

$$= \frac{8 \times 9}{2} = 36$$

18. (d) $I = \int_0^\pi [\cot x] dx$

$$I = \int_0^\pi [\cot(\pi - x)] dx$$

$$I = \int_0^\pi [-\cot x] dx$$

Add Eqs. (i) and (ii), $2I = \int_0^\pi ([\cot x] + [-\cot x]) dx$

Using result $t[x] + [-x] = \begin{cases} 0 & \text{if } x \in I \\ -1 & \text{if } x \notin I \end{cases}$

$$= \int_0^\pi -1 dx$$

$$2I = -[x]_0^\pi \Rightarrow 2I = -\pi \Rightarrow I = -\frac{\pi}{2}$$

19. (b) It is an even function

$$\int_{-\pi/3}^{\pi/3} \frac{x \sin x}{\cos^2 x} dx = 2 \int_0^{\pi/3} \frac{x \sin x}{\cos^2 x} dx = 2 \int_0^{\pi/3} x \tan x \sec x dx$$

Using integration by parts

$$2 \left([x \sec x]_0^{\pi/3} - \int_0^{\pi/3} 1 \times \sec x dx \right)$$

We should know, $\int \sec x dx = \log |\sec x + \tan x|$

$$= 2 \left[\left(\frac{\pi}{3} \sec \frac{\pi}{3} - 0 \right) - \log \left(\sec \frac{\pi}{3} + \tan \frac{\pi}{3} \right) \right]$$

$$+ \log(\sec 0 + \tan 0)$$

$$= 2 \left(\frac{\pi}{3} \times 2 \right) - 2(\log(2 + \sqrt{3}))$$

We know that $\tan \frac{5\pi}{12} = 2 + \sqrt{3} = \frac{4\pi}{3} - 2 \log \tan \left(\frac{5\pi}{12} \right)$

20. (a) $\int_0^\pi x^3 \sin x dx$

Using integration by parts

$$= x^3 [-\cos x] - \int (-\cos x) \cdot 3x^2 dx$$

$$= -x^3 \cos x + 3 \int x^2 \cos x dx$$

$$= -x^3 \cos x + 3[x^2(\sin x) - \int \sin x \cdot (2x) dx]$$

$$= -x^3 \cos x + 3x^2 \sin x - 6 \int x \sin x dx$$

$$= -x^3 \cos x + 3x^2 \sin x - 6[x(-\cos x) - \int (-\cos x) \cdot 1 dx]$$

$$= -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x$$

$$\therefore \int_0^\pi x^3 \sin x dx = [-x^3 \cos x$$

$$+ 3x^2 \sin x + 6x \cos x - 6 \sin x]_0^\pi$$

$$= -\pi^3 \cos \pi + 3\pi^2 \sin \pi + 6\pi \cos \pi - 6 \sin \pi = \pi^3 - 6\pi$$

21. (b) $\int_0^1 x(1-x)^n dx = \int_0^1 (1-x)(1-(1-x))^n dx$

$$= \int_0^1 (1-x)x^n dx$$

$$= \left[\frac{x^{n+1}}{n+1} \right]_0^1 - \left[\frac{x^{n+2}}{n+2} \right]_0^1 = \frac{1}{n+1} - \frac{1}{n+2}$$

$$= \frac{n+2-n-1}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}$$

22. (a) $f(x) = \sin^5 x + \sin^3 x \Rightarrow f(-x) = -\sin^5 x - \sin^3 x$

$$f(x) + f(-x) = \sin^5 x + \sin^3 x - \sin^5 x - \sin^3 x = 0$$

$$g(x) = \cos^5 x + \sin^3 x \Rightarrow g(-x) = \cos^5 x - \sin^3 x$$

$$g(x) + g(-x) = 2\cos^5 x$$

Then, find $\int_0^{\pi/2} [f(x) + f(-x)][g(x) + g(-x)] dx$

$$= \int_0^{\pi/2} 0 \times 2\cos^5 x dx = 0$$

23. (b) $I = \int_0^\pi x f(\sin x) dx$

$$I = \int_0^\pi (\pi - x) f(\sin(\pi - x)) dx$$

$$I = \int_0^\pi (\pi - x) f(\sin x) dx$$

$$I = \pi \int_0^\pi f(\sin x) dx - \int_0^\pi x f(\sin x) dx$$

Add Eqs. (i) and (ii),

$$2I = \pi \int_0^\pi f(\sin x) dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^\pi f(\sin x) dx$$

24. (c) Let $f: R \rightarrow R$ be defined by $f(x) = \begin{cases} x+2, & \text{if } x < 0 \\ |x-2|, & \text{if } x \geq 0 \end{cases}$

$$\int_{-2}^0 (x+2) dx + \int_0^2 (2-x) dx + \int_2^3 (x-2) dx$$

$$= \left[\frac{x^2}{2} \right]_{-2}^0 + 2[x]_{-2}^0 + 2[x]_0^2 - \left[\frac{x^2}{2} \right]_2^3 + \left[\frac{x^2}{2} \right]_2^3 - 2[x]_2^3 = 4.5$$

25. (b) $\int_{\log 2}^x \frac{1}{\sqrt{e^x-1}} dx = \frac{\pi}{6} e^x - 1 = t^2 \Rightarrow e^x = t^2 + 1$

$$\Rightarrow e^x dx = 2t dt$$

$$= \int \frac{2t dt}{(t^2+1)t}$$

$$\Rightarrow dx = \frac{2t dt}{e^x}$$

$$= 2 \int \frac{dt}{t^2+1} \Rightarrow dx = \frac{2t}{t^2+1} dt$$

$$= 2 \tan^{-1} t$$

$$[2 \tan^{-1} \sqrt{e^x-1}]_{\log 2}^x = \frac{\pi}{6}$$

$$\Rightarrow \tan^{-1} \sqrt{e^x-1} - \tan^{-1} \sqrt{e^{\log 2}-1} = \frac{\pi}{12}$$

$$\Rightarrow \tan^{-1} \sqrt{e^x-1} - \tan^{-1} 1 = \frac{\pi}{12}$$

$$\Rightarrow \tan^{-1} \sqrt{e^x-1} - \frac{\pi}{4} = \frac{\pi}{12}$$

$$\tan^{-1} \sqrt{e^x-1} = \frac{\pi}{3}$$

$$\Rightarrow \sqrt{e^x-1} = \tan\left(\frac{\pi}{3}\right)$$

$$\Rightarrow \sqrt{e^x-1} = \sqrt{3}$$

Squaring on both side,

$$e^x - 1 = 3$$

$$\Rightarrow e^x = 4$$

$$\log e^x = \log 4$$

$$\Rightarrow x \log e = \log(2)^2 = 2 \log 2$$

$$\Rightarrow x = 2 \log 2$$

Taking log on both sides, we get



$$\log e^x = \log 4$$

$$\Rightarrow x \log e = \log(2)^2 = 2 \log 2$$

$$\Rightarrow x = 2 \log 2$$

26. (a) $I_n = \int_0^a (a^2 - x^2)^n \cdot 1 dx$

Using integration by parts

$$I_n = [(a^2 - x^2)^n \cdot x]_0^a - \int_0^a n(a^2 - x^2)^{n-1}(-2x)xdx$$

$$I_n = 0 + 2n \int_0^a (a^2 - x^2)^{n-1} x^2 dx$$

$$I_n = -2n \int_0^a (a^2 - x^2)^{n-1} (a^2 - x^2 - a^2) dx$$

$$I_n = -2n \int_0^a (a^2 - x^2)^n + 2na^2 \int_0^a (a^2 - x^2)^{n-1} dx$$

$$I_n = -2nI_n + 2na^2 I_{n-1}$$

$$\Rightarrow I_n = \frac{2na^2}{2n+1} I_{n-1}$$

27. (b) $\int_{-2}^2 (ax^5 + bx^3 + c) dx$

$$\int_{-2}^2 ax^5 + \int_{-2}^2 bx^3 + c \int_{-2}^2 dx = 0 + 0 + c \int_{-2}^2 dx$$

$$= c \int_{-2}^2 dx = 4c$$

depends on c .

28. (d) $\int_a^b |x-a| dx + \int_a^b |x-b| dx$ as $a < x < b$

$$= \int_a^b (x-a) dx - \int_a^b (x-b) dx$$

$$= b \int_a^b dx - a \int_a^b dx$$

$$= (b-a)^2$$

29. (d) $\lim_{n \rightarrow \infty} \frac{1^a + 2^a + \dots + n^a}{(n+1)^{a-1} \{ (na+1) + (na+2) + \dots + (na+n) \}} = \frac{1}{60}$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^a}{(n+1)^{a-1} [n^2 a + \frac{n(n+1)}{2}]} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{2 \sum_{r=1}^n \left(\frac{r}{n}\right)^a}{\left(1+\frac{1}{n}\right)^{a-1} \cdot (2na+n+1)} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} \left(2 \sum_{r=1}^n \left(\frac{r}{n}\right)^a \right) \lim_{n \rightarrow \infty} \frac{1}{\left(1+\frac{1}{n}\right)^{a-1} \cdot \left(2a+1+\frac{1}{n}\right)} = \frac{1}{60}$$

$$= 2 \int_0^1 (x^a) dx \cdot \frac{1}{1 \cdot (2a+1)} = \frac{1}{60} = \frac{2 \cdot [x^{a+1}]_0^1}{(2a+1) \cdot (a+1)} = \frac{1}{60}$$

$$= 2 \int_0^1 (x^a) dx \cdot \frac{1}{1 \cdot (2a+1)} = \frac{1}{60} = \frac{2 \cdot [x^{a+1}]_0^1}{(2a+1) \cdot (a+1)} = \frac{1}{60}$$

$$\therefore \frac{2}{(2a+1)(a+1)} = \frac{1}{60}$$

$$\Rightarrow (2a+1)(a+1) = 120$$

$$\Rightarrow 2a^2 + 3a + 1 - 120 = 0$$

$$\Rightarrow 2a^2 + 3a - 119 = 0$$

$$\Rightarrow (2a+17)(a-7) = 0$$

$$\Rightarrow a = 7, \frac{-17}{2}$$

30 (b) Refer to solution 16.

31. (d)

32. (c)

Given:

$$\frac{d}{dx} \int_{\sin x^2}^{\sin x} e^{t^2} dt \Big|_{z=x}$$

Using Leibniz's Rule:

$$= e^{(\sin x)^2} \cdot 2 \cos x - e^{(\sin x^2)^2} \cdot 2x \cos(x^2)$$

At $x = \pi$.

$$- \sin \pi = 0, \text{ so first term} = e^0 \cdot 2(-1) = -2$$

- Second term ≈ 0 (since $\cos(\pi^2)$ is small)

Ans. -2

33. (b) $\int_a^3 e^{x^2} dx = \int_a^5 e^{x^2} dx + \int_5^3 e^{x^2} dx$

$$\int_a^b f(x) = \int_a^c f(x) + \int_c^b f(x)$$

Option (b) is correct.

34. (b)

$$cf(x) + d f(1/x) = |\log |x|| + 3$$

$$\int_1^e f(x) dx = ?$$

$$c^2 f(x) + cd f(1/x) = c |\log |x|| + 3c \dots\dots\dots(i)$$

$$d^2 f(x) + cd f(1/x) = d |\log |x|| + 3d \dots\dots\dots(ii)$$

Equation (i) and (ii) subtract

$$(c^2 - d^2) f(x) = (c-d) |\log |x|| + 3(c-d)$$

$$(c^2 - d^2) f(x) = (c-d)(|\log |x|| + 3)$$

$$f(x) = \frac{c-d}{c^2-d^2} (|\log |x|| + 3)$$

$$\frac{c-d}{c^2-d^2} \int_1^e |\log |x|| + 3$$

$$\frac{c-d}{c^2-d^2} \int_1^e \log x + 3 = \frac{c-d}{c^2-d^2} [x \log x - x + 3x]_1^e$$

$$= \frac{(c-d)(3e-2)}{c^2-d^2}$$